Thus,

$$\mathrm{Tr}\,|_{S^2V}(A\otimes A)-\mathrm{Tr}\,|_{\Lambda^2V}(A\otimes A)=\sum_{1\le i\le n}\lambda_i^2=\mathrm{Tr}(A^2).$$

Thus for $g\in G$ we have

$$\chi_V(g^2)=\chi_{S^2V}(g)-\chi_{\Lambda^2V}(g).$$

Therefore, setting $P=|G|^{-1}\sum_{g\in G}g$, we get

$$|G|^{-1}\chi_V(\sum_{g\in G}g^2)=\chi_{S^2V}(P)-\chi_{\wedge^2V}(P)=\dim(S^2V)^G-\dim(\wedge^2V)^G$$

$$=\begin{cases}1 & \text{if } V \text{ is of real type,}\\ -1 & \text{if } V \text{ is of quaternionic type,}\\ 0 & \text{if } V \text{ is of complex type.}\end{cases}$$

Finally, the number of involutions in $G$ equals

$$\frac{1}{|G|}\sum_V \dim V\chi_V(\sum_{g\in G}g^2)=\sum_{\text{real } V}\dim V-\sum_{\text{quat. } V}\dim V.$$

□

**Corollary 5.1.6.** *Assume that all representations of a finite group $G$ are defined over real numbers (i.e., all complex representations of $G$ are obtained by complexifying real representations). Then the sum of the dimensions of all the irreducible representations of $G$ equals the number of involutions in $G$.*

**Exercise 5.1.7.** Show that any nontrivial finite group of odd order has an irreducible representation which is not defined over $\mathbb{R}$ (i.e., not realizable by real matrices).

## 5.2. Algebraic numbers and algebraic integers

We are now passing to deeper results in the representation theory of finite groups. These results require the theory of algebraic numbers, which we will now briefly review.

**Definition 5.2.1.** $z\in\mathbb{C}$ is an **algebraic number** (respectively, an **algebraic integer** ) if $z$ is a root of a monic polynomial with rational (respectively, integer) coefficients.